Males and Females (Machine) Learning to Express Emotions More Granularly

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Abstract

As Americans report heightened levels of depression, anxiety, and stress, the need to identify underlying predictors of mental health is increasingly important. Identifying one's emotions with granular detail is associated with down regulation of negative emotions. However, the frequency and intensity one experiences these mental health indicators differs by gender, and the individual capacity to identify emotions is highly variable. Existing research has explored gender differences in diagnoses and behavioral affect of females and males with moderate to severe depression, anxiety, and stress, but the relative importance of features predicting these mental health indicators by gender is under explored. This study leveraged machine learning algorithms to identify gender differences in important features predictive of mental health in a sample of 38,835 participants. Results revealed the relative importance of anxiety and stress features were similar for females and males. By contrast, female depression was better predicted by anxiety and stress features, while male depression was best predicted by features associated with depression. Similarly, personality was the strongest predictor of female depression, anxiety, and stress, while male stress was better predicted by features such as family size and sexual orientation. Finally, emotional stability was more important for males across all three mental health indicators than for females. These results highlight the need to equip females and males with more granular emotional language to parse their experiences with, and sources of, depression, anxiety, and stress.

"When we can talk about our feelings, they become less overwhelming, less upsetting, and less scary." – Fred Rogers (1994, p. 115)

Whether the advice comes from a beloved neighbor like Mr. Rogers or emotional intelligence researchers Peter Salovey and John D. Mayer (1990), the ability to identify and regulate emotions is essential for developing a healthy personality, cultivating friendships, and enacting social change.

As Americans report worsening mental health (Jones, 2025), elevated levels of depression (Witters, 2026), heightened anxiety (American Psychiatric Association, 2026), and the highest levels of stress in decades (Fioroni & Foy, 2024), the need to identify physiological and behavioral symptoms of mental health indicators such as depression, anxiety, and stress is of increasing importance. After all, as Mr. Rogers put it, "Anything that's human is mentionable, and anything that is mentionable can be more manageable" (1994, p. 115).

Indeed, labeling emotions is associated with regulation of distress and other negative affect emotions (Kashdan et al., 2015; Kircanski et al., 2012; Levy-Gigi & Shamay-Tsoory, 2022; Yue et al., 2025). Problematically, research by authors Bradberry and Greaves (2009) revealed that only 36% of people could identify their emotions when they experienced them (p. 19). Nook et al. (2020) found that while emotional vocabulary increased during childhood, it plateaued between 11 and 18 years of age.

Given the importance of identifying emotions in developing healthy emotional regulation (Mayer & Salovey, 1995; Salovey & Mayer, 1990), the gap between emotional experience and the granularity with which a person can verbalize that experience has become a field of study in psychology.

Emotional Granularity

The term emotional granularity was coined by researcher Lisa Barrett (2004) to describe the level of verbal precision with which a person identified emotions they experienced. An individual who rates low in emotional granularity is said to report their experiences in general terms as positively or negatively valenced. An individual who rates high in emotional granularity is said to report to their emotional experiences in more descriptive terms (e.g., happy, sad, angry) such that a distinct feeling is conveyed.

The ability to distinctly describe one's emotions is associated healthier coping mechanisms and better mental health (Kashdan et al., 2015; Smidt & Suvak, 2015). For example, individuals who rate high on emotional granularity are less prone to maladaptive behavior like hitting or binge eating (Dixon-Gordon et al., 2014).

Furthermore, individuals with high emotional granularity self-report more frequent emotional regulation (Barrett et al., 2001) and increased resilience (Tugade et al., 2004), while individuals low on emotional granularity tend to be less successful in downregulating negative emotions (Kalokerinos et al., 2019).

Taken together, the connection between experiencing and explaining is beneficial to wellbeing such that providing people with the vocabulary clinicians use to describe the physiological and behavioral predictors of depression, anxiety, and stress is a necessary undertaking.

Gender Differences

The extent to which individuals experience indicators of depression, anxiety, and stress can depend on gender. For instance, females experience a higher lifetime prevalence and diagnosis of anxiety disorders (Kessler et al., 2005; McLean et al., 2011) and score higher on scales measuring stress (Graves et al., 2021; Matud, 2004) compared to males.

Additionally, females rate lower on personality trait emotional stability (Lippa, 2010; Weisberg et al., 2011), and although personality tends to be a stronger predictor of depression, anxiety, and stress for males, emotional stability is the best personality predictor for depression, anxiety and stress for both males and females (Alizadeh et al., 2018; Uliaszek et al., 2010).

Evidence also suggests that females rely on more emotion-focused coping styles for stress compared to males (Matud, 2004), and in response to depression, females are more prone to internalizing behavior (Levkovich et al., 2025; Li et al., 2023), and rumination (Johnson & Whisman, 2013).

To date, much of the research on gender differences in experiencing depression, anxiety, and stress has focused either on the magnitude of the differences (Salk et al., 2017), the behavioral expressions of individuals experiencing high levels of one or the other (Kuehner, 2017; Sloan & Sandt, 2006), or associated factors such as biology (Donner & Lowry, 2013; Farhane-Medina et al., 2022).

For example, a widely cited study by Martin et al. (2013) found that although the proportion of females and males who experience depression is about the same (e.g., 33.3% vs. 30.6%, respectively), males reported higher rates of anger, aggression, substance abuse, and risk taking. Conversely, females tend to report depressed mood, appetite and sleep disturbance with higher frequency and intensity compared to males (Cavanagh et al., 2017).

While there is a broad swath of research on gender differences in symptomatic expression, a topic that remains under explored is whether females and males differ in how depression, anxiety, and stress are experienced in self-reported measures (e.g., does dysphoria better predict depression for females while hopelessness better predicts depression for males?). Given gender differences in diagnoses and symptomatic expression, it is plausible the underlying subscales of measures for depression, anxiety, and stress would differently predict symptoms for females and males.

DASS Root Vocabulary

The 21-item Depression Anxiety Stress Scales (DASS-21) is a widely used and rigorously validated psychometric tool that operationally defines depression, stress, and anxiety in terms of clinically aligned physiological and behavioral subscales (Adu et al., 2025; Antony et al., 1998; Dwight et al., 2024; Henry & Crawford, 2005; P. F. Lovibond & Lovibond, 1995; S. H. Lovibond & Lovibond, 1995).

Researchers have demonstrated population validity in samples of youths (Dwight et al., 2024), across cultures (Adu et al., 2025; Henry & Crawford, 2005; P. F. Lovibond & Lovibond, 1995), and between the sexes (Gomez et al., 2014). Therefore, the DASS-21 subscales are an ideal source of clinically aligned, emotionally granular vocabulary.

The subscales of DASS-21 describe physiological and behavioral predictors. For example, depression is characterized by hopelessness (i.e., nothing to look forward to), while anxiety is characterized by autonomic arousal such as dryness of mouth or difficulty breathing (Antony et al., 1998; P. F. Lovibond & Lovibond, 1995). Table 1 shows the DASS-21 scales with the associated subscales, items, and root vocabulary. Parsing these predictors and properly labelling them should, in theory, add to an individual's capacity for emotional granularity, yielding all the associated benefits.

Present Study

This study proposes applying machine learning models to the Answers to the Depression Anxiety Stress Scales data set available from the Open Psychometrics Project (2019). In a developing study by Bradley Allen (2026), confirmatory factor analysis was conducted on this dataset confirming the DASS-21 measure functioned the same for both females and males, allowing for comparisons between genders.

The data further replicated findings of gender differences in personality, personality as a predictor of depression, anxiety, and stress, as well as findings that females tended to experience higher levels of all three mental health indicators compared to males. Therefore, the Answers to the Depression Anxiety Stress Scales data set is a strong candidate for generalizable findings of predictors of depression, anxiety, and stress.

Existing machine learning research has analyzed this data set to predict depression, anxiety, and stress using demographic variables such as family size, gender, and race (Kumar et al., 2020; Yesudas, 2023). Other applied machine learning research has analyzed physiological predictors of depression, anxiety, and stress such as skin conductance and heart rate (Al Abdi et al., 2026; Liu et al., 2025). A metanalysis of models predicting depression, anxiety, and stress revealed the most common models were kNN, SVM, random forest, and ensemble (Taskynbayeva & Gutoreva, 2025).

Despite the extensive research on this topic, whether subscales of the DASS-21 are better predictors for depression, anxiety, and stress for females and males remains under explored. This project aims to extend research on this topic in two ways. First, by applying newer models such as XGBoost, LightGBM, and CatBoost. Second, by using these models in a novel way to predict moderate to severe levels of depression, anxiety, and stress for males and females.

These findings will help males and females in multiple ways. First, if gender differences in physiological and behavioral predictors exist, these findings will help males and females identify symptomatic experiences of depression, anxiety, or stress separately. Properly identifying symptoms will help males and females communicate the feelings they experience to one another, as well as enhance interactions with mental health care professionals by using more precise language.

Method

Participants

The raw data contained 39,775 responses. Participants were required to pass a series of checks for their responses to be included. First, the survey included a 16-word list (e.g., boat, paucity, lucid), 3 of which were fake (e.g., cuivocal, florted, verdid). Participants were instructed to check a box next to words whose definitions they knew for certain. Those who indicated they knew the definitions of all 3 fake words were excluded from the analysis, resulting in a reduction of 326 participants.

Second, participants were required identified as male or female. As a result, 540 participants who selected “other” and 67 participants who did not provide a response were excluded. Finally, 7 participants who listed their age as greater than 100 were removed. The remaining sample size was 38,835.

Measures

Depression, Anxiety, and Stress

Depression, anxiety, and stress scores were calculated using the 21-item Depression Anxiety Stress Scale (DASS-21; P. F. Lovibond & Lovibond, 1995). Depression was assessed using a 7-item scale that included items such as “I felt sad and depressed” and “I just couldn’t seem to get going.” Anxiety was assessed using a 7-item scale that included items such as “I had difficulty swallowing” and “I felt terrified.” Stress was assessed using a 7-item scale that included items such as “I found it hard to wind down” and “I tended to over-react to situations.”

Items for all three scales were measured using a 4-point Likert scale (0 = *Did not apply to me at all*, 3 = *Applied to me very much, or most of the time*). Scores were then doubled for the DASS-21 subset (S. H. Lovibond & Lovibond, 1995). Descriptive statistics are shown in Table 2.

Personality

Personality scores were calculated using the Ten Item Personality Inventory (TIPI; Gosling et al., 2003). Participants responded to the prompt “I see myself as” and rated items such as “Extraverted, enthusiastic” and “Calm, emotionally stable.” Extraversion, agreeableness, conscientiousness, emotional stability, and openness to experience were all assessed using a 2-item scale, on a 7-point Likert scale (1 = *Disagree strongly*, 7 = *Agree strongly*).

Openness to experience measures intellectual curiosity, creativity, and the preference of novelty. Conscientiousness measures self-discipline, orderliness, and persistence toward goals. Extraversion measures traits such as energy, assertiveness, sociability, and the tendency to seek stimulation in the company of others. Agreeableness measures a person’s tendency to be compassionate, cooperative, and seek social harmony. Emotional stability is the tendency to regulate negative emotions and exhibit low reactivity to stress, anxiety, and environmental pressure.

Procedure

The Open Psychometrics (2019) data set Answers to the Depression Anxiety Stress Scales was explored. Observations included in the data were gathered through an online survey from 2017 to 2019 and recorded 39,775 responses. The raw data frame contained 172 variables, including 42 items for the Depression Anxiety Stress Scales (DASS; P. F. Lovibond & Lovibond, 1995), 10 items from the Ten Item Personality Inventory (TIPI; Gosling et al., 2003), 16 items for the attention check (i.e., one variable for each word), and a series of time related questions (e.g., how much time spent on each question, total survey completion time) and demographic variables (e.g., education level, college major, gender, religion, orientation).

Data Preprocessing and Analytic Strategy

Software

To analyze the data, 7 models were trained and tested using Scikit-learn (Pedregosa et al., n.d.) with Python (Python Software Foundation, 2025) in Visual Studio Code (Microsoft, 2026). Models included Random Forest (Breiman, 2001), AdaBoost (Freund & Schapire, 1997), Gradient Boosting (Friedman, 2001), Histogram Gradient Boosting (Friedman, 2001), XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018). Additional software packages used in this project were matplotlib (Hunter, 2007), numpy (Harris et al., 2020), pandas (The Pandas Development Team, 2020), and Seaborn (Waskom, 2021).

DASS-21 and demographic plots contrasting feature importance by gender were created in R Studio (Posit team, 2026) using R (R Core Team, 2026) libraries patchwork (Pedersen, 2025), tidytext (Robinson et al., 2026), and tidyverse (Wickham et al., 2019).

Model Design

The data was filtered into two data frames, one for males and one for females. The data frames were then split into one containing DASS-21 variables, and another containing demographic variables. These data frames were used to identify the top DASS-21 subscales and demographic predictors of mental health (i.e., depression, anxiety, and stress) for males and females, respectively.

In observance of cost-sensitive machine learning practices for applied psychology, Randomized Search CV was applied to optimize model parameter tuning (Feng et al., 2026; Sterner et al., 2025). Parameters included 20 n-iterations, 5-fold cross validation, and accuracy as the scoring metric. Models included "honesty" provisions such as minimum sample splits, variable tree depth, and weighting described by Athey et al. (2019). Given the large sample size, minimum sample splits were set to 30 to ensure that each node generalized to the population under the Central Limit Theorem.

Finally, the best model parameters were extracted, then applied to an Ensemble model. Models were trained on 80% of the data and tested on 20%. In all cases, random state was set to 42 for replicability.

The Ensemble model was used to find the best predictors of depression, anxiety, and stress for males and females, respectively. Soft voting was applied for its ability to capture the nuances of complex, multi-dimensional mental health data when using Ensemble with models such as decision trees, XGBoost, CatBoost, and LightGBM (Gupta & Pandit, 2026; Jain et al., 2025; Pavani et al., 2024). Data preprocessing was required before models were trained.

DASS-21 Clinical Cut-off Scores

To facilitate classification, binary labels were created from established clinical cut-off scores by S.H. Lovibond and Lovibond (1995). Using the DASS-21 measure, scores within the normal to mild range were assigned to 0, while scores within the moderate to extremely severe range were assigned to 1. Score ranges are shown in Table 3.

Before the scores were calculated, respondent values needed to be adjusted downward by 1, as the original scale range was 0-3, but the survey calculated them from 1-4. This brought the values in line with the DASS-21 clinical cut-off ranges (P. F. Lovibond & Lovibond, 1995; S. H. Lovibond & Lovibond, 1995; Soria-Reyes et al., 2024).

Personality Reverse Scoring

The TIPI items were assessed using a 7-point Likert scale, which included 5 reverse-scored items (i.e., questions 2, 4, 6, 8, and 10). Reversed scored items were calculated by subtracting each response from 8 (e.g., 8 - 1, 8 - 2, etc.) before taking the mean of each set and assigning them to their respective trait (Gosling et al., 2003).

Grouped Categories

To address model bias in decision trees, religion and age groups were assigned numerical categories. These variables contained multiple categories or high ranges, which decision trees favor as important features to split on (Nembrini et al., 2018; Strobl et al., 2007). For instance, age ranged from 13 to 89 (Table 4), and religion contained 12 categories. Meaningful age groups (Geifman et al., 2013) were assigned as follows: 1 = *Adolescent*, 2 = *Young adult*, 3 = *Adult*, 4 = *Middle aged*, 5 = *Aged*, 6 = *80 and over* (Table 2).

Religion was converted to a binary variable (0 = *Not religious*, 1 = *Religious*). Participants who selected agnostic and atheist were assigned to the not religious group, while participants who selected Buddhist, Christian, Hindu, Jewish, Muslim, Sikh, and other were assign to the religious group (Table 2).

Casting Categorical Variables to Data Type Categorical

Categorical variables in machine learning have been known to contribute to model bias in machine learning through spurious mathematical correlation (Kopp et al., 2025; Zhu et al., 2024). To eliminate a potential source of bias, ordered categorical variables were cast to type category using Pandas astype (Pandas via NumFOCUS, Inc., 2023).

Education was ordered by level of attainment (e.g., 1 = *Less than high school*, 4 = *Graduate degree*). Age group was ordered by epoch (1 = *Adolescent*, 6 = *80 and over*). The remainder of categorical variables were unordered (i.e., urban, gender, orientation, race, voted, married).

DASS-21 and TIPI items were either sums or averages, respectively, and thus treated as continuous numerical variables (Gosling et al., 2003; P. F. Lovibond & Lovibond, 1995).

Class Imbalance

After applying the binary classifier to participant DASS scores, notable class imbalances were observed. Differences were more common for females, with 30.65%, 29.39%, and 38.56% experiencing low to mild symptoms of depression, anxiety, and stress, compared to 69.35%, 70.61%, and 61.44% experiencing moderate to severe symptoms, respectively (Table 2).

The proportion of males with anxiety and stress was approximately equal at 41.89% and 52.62% experiencing low to mild symptoms, compared to 58.11% and 47.38% experiencing moderate to severe symptoms. The largest difference was on depression, with 34.97% of males experiencing low to mild symptoms, compared to 65.03 experiencing moderate to severe symptoms (Table 2).

To address class imbalance, class weight was set to balanced for Random Forest, LightGBM, and CatBoost. A custom weight was applied to XGBoost by multiplying by the proportion of negative to positive cases (i.e., low to mild divided by moderate to high). Weights were not applied to AdaBoost, Gradient Boosting, or HistGradientBoost.

Missing Data

No patterns of missing data were found for DASS-21, TIPI, gender, or other demographic variables.

Evaluation

Model performance was evaluated using accuracy, precision, recall, and F1 score. Accuracy measures the overall percentage of correct predictions, calculated as the ratio of true results (both positive and negative) to all total outcomes. Precision evaluates the reliability of the model's positive predictions—such as identifying cases of moderate-to-severe depression—by calculating the ratio of true positives to all predicted positives. Recall assesses the model's capacity to capture all actual positive cases by calculating the ratio of true positives to all true positive and false negative outcomes. F1 score is the ratio of the product of precision and recall over the sum of precision and recall, and is used as a balanced summary metric to indicate whether the model was over optimized for one metric over the other (Sujon et al., 2025; Yesudas, 2023)

Results

DASS-21 Subscales as Predictors of Depression, Anxiety, and Stress

Model Performance

As expected, the baseline models performed with nearly 100% accuracy. This is the case because the subscales should accurately predict the scale (e.g., the subscales aggregated for the depression score should accurately predict depression). The Ensemble model performed equally well for males and females, with universal accuracy at or above 99% (Table 5). Optimized parameter values and model performance contrasts by gender are included in the Supplement (Tables S1, S2, S3; Figs. S1, S2, S3).

Allowing model parameters to freely vary between females and males resulted in average differences in model parameters. For example, to predict anxiety, AdaBoost models were optimized for females vs. males with 1000 vs 500 n-estimators, .10 vs .50 learning rate, and entropy vs. Gini criterion (Table S2). This suggests that Random Grid Search CV was useful in tuning cost-sensitive models (Sterner et al., 2025).

Feature Importance

Feature importance was calculated and extracted from the Ensemble model. The values were standardized to allow between group comparisons by dividing each feature score by the highest feature score. The result is a relative feature score where each is a fraction of the highest feature score (Table 6).

The feature scores for anxiety and stress performed as expected for both females and males, with important features strictly grouping under the associated scale. Anxiety, for example, had a top feature of autonomic arousal, followed by subjective experience of anxious affect, situational anxiety, and skeletal musculature effects, all of which ranked 74% or higher for females and 80% or higher for males (Fig. 3). Conversely, all other features ranked 2% or less for females, and 5% or less for males, indicating strong discriminant validity. The order of importance was approximately the same for both females and males.

Stress followed the same pattern, with a top feature of easily upset, followed by irritable / over-reactive, impatient, nervous arousal, and difficulty relaxing. For females, the stress features ranged from 68% to 90% of the top feature, and 70% to 92% for males (Table 6). The order of importance was approximately the same (Fig. 1). All other features ranked 11% or lower for females, and 27% or lower for males. By contrast, feature importance of depression trended higher across all features, with a minimum relative feature importance of 47% for females, and 35% for males (Table 6). Additionally, there were no shared features between females and males in the top 5 important features (Table 6).

Top features predictive of depression for females included easily upset, impatient, irritable / over-reactive, autonomic arousal, and subjective experience of anxious affect. Conversely, the top 5 features for males were inertia, hopelessness, self-deprecation, lack of interest / involvement, and dysphoria. See Fig. 2 for chart of top features in ascending order by gender.

Interestingly, the top 5 features for females included a mixture of anxiety and stress subscales but none for depression, while the top 5 features for males were strictly subscales associated with depression (Table 1; Fig. 2). This suggests that for clinical diagnoses, DASS-21 depression subscales perform differently in machine learning models for females experiencing moderate to severe depression when compared to males.

Demographic Information as Predictors of Depression, Anxiety, and Stress

Model Performance

Baseline models demonstrated variable performance for females and males, with accuracy fluctuating between 70% to 76% for depression (Table S4; Fig. S4), anxiety (Table S5; Fig. S5), and stress (Table S6; Fig. S6). On balance, the Esemble models performed higher than the baseline for females and males with depression at 76% for both, anxiety at 74% and 72%, and stress at 75% and 74% accuracy, respectively (Table 7). Given the extent to which demographic variables may only be tangentially related to depression, anxiety, and stress, these rates should be expected to trend lower than the subscales of DASS-21. Confusion matrices revealed that demographic variables were between 20 and 30% less accurate predicting negative cases of depression, anxiety, and stress compared to positive cases (Figs. S7, 8, 9). This relation held for all models except male stress, where accuracy was 73% in both cases.

Important Features

Personality tended to be the best predictor of depression, and anxiety for both females and males. Across depression, anxiety and stress, the top three predictors for females were extraversion, conscientiousness, and agreeableness, all of which ranked 80% or higher compared to the top feature (Table 8). For anxiety and stress, the next top predictors were family size and openness to experience, while emotional stability, then family size were the next best predictors for depression (Table 8; Fig. 3).

Similarly, personality was also the top predictor of depression and anxiety for males. Depression was best predicted by extraversion, conscientiousness, emotional stability, openness to experience, then family size. Anxiety was best predicted by extraversion, emotional stability, conscientiousness, agreeableness, and family size (Table 8; Fig. 3). Notably, compared to females, the relative importance of each subsequent feature from the main feature for males was 3% to 20% lower. This suggests that a broader range of features elicits depression, anxiety, and stress in females compared to males.

Conversely, the top features indicative of moderate to severe stress in males differed significantly. Compared to anxiety and depression, the top feature for males was emotional stability, followed by family size, sexual orientation, agreeableness, and education level (Fig. 4). In total, personality was not as predictive of stress for males.

Discussion

Ensemble machine learning revealed distinct patterns in relative feature importance for females and males who reported moderate to severe levels of depression, anxiety, and stress. For instance, male depression was best predicted by depression features. Contrary to expectations, depression feature importance scores trended higher for females, and the top predictive features for female depression were more similar to anxiety and stress.

Interrelated important features sheds some light into why females might experience higher lifetime diagnoses of anxiety disorders and score higher on scales measuring stress compared to males (Graves et al., 2021; Kessler et al., 2005; Matud, 2004; McLean et al., 2011). It further suggests that what it means to experience depression for females may differ from males in clinically important ways. For instance, it may be the case the females are better treated for underlying anxiety and stress disorders in tandem with depression, while males may be treated strictly for depression.

With respect to important demographic features, results supported findings that personality tends to be a strong predictor of depression, anxiety, and stress. However, this relationship is slightly different for males compared to females with respect to stress. Feature importance revealed that emotional stability, followed by family size, sexual orientation, agreeableness, and education level, all of which rated 50% or higher in importance compared to emotional stability, mattered more than openness to experience, conscientiousness, and extraversion.

While emotional stability has been found to exert the most influence over depression, anxiety, and stress for both females and males compared to other personality traits (Alizadeh et al., 2018; Uliaszek et al., 2010), the pathway to emotional stability as an important predictor looks different for females compared to males. These results suggest that for females, the pathway to emotional stability runs through extraversion and conscientiousness, with agreeableness playing a significant role along the way. By contrast, for males, emotional stability was the primary feature for stress, second for anxiety, and third for depression.

This is significant because females tend to rate higher in trait agreeableness and lower in trait emotional stability (Lippa, 2010; Weisberg et al., 2011), findings which have been replicated in this dataset (Allen, 2026). Thus, higher relative feature importance of these traits for females who tend rate higher in agreeableness and lower in emotional stability further sheds light into lifetime difference in clinical diagnoses of depression, anxiety, and stress. For males, a higher overall relative feature importance of emotional stability suggests emotional stability drives symptoms of mental health for males who suffer from moderate to severe depression, anxiety, and stress.

While these findings do not depart from convention, they offer a sense of ordinality to the experiences and traits that most affect mental health for females and males separately. These results are supported by rigorously designed machine models including cost sensitivity (Feng et al., 2026; Sterner et al., 2025), population-level generalizable minimum sample splits (i.e., no less than 30 participants per node), and model honesty provisions (Athey et al., 2019).

Ordinal feature importance suggests learning the language clinicians use to diagnose depression, anxiety, and stress would help males and females in different ways. For females, placing a strong emphasis on language describing anxiety and stress, and how to parse them from depression would improve the granularity with which symptoms are described and managed. For males, similar attention should be paid to descriptors of anxiety and stress, but also on depression. Pairing this language with understanding how personality differently affects experiencing depression, anxiety, and stress would equip females and males alike with more tools to identify sources associated with declines in mental health.

Limitations and Future Directions

The data was collected via convenience sampling. Convenience sampling is known to introduce bias such as social desirability and halo effect, especially in survey on desirable traits (Speklé & Widener, 2018). While attempts were made to ensure response integrity, such as removing participants who failed the attention check, and participants who listed their age as 100 years or more, the need for a large sample size for the project prohibited more stringent methodological constraints such as down sampling to mimic a random sample from the responses. It is also likely that the sample underrepresents low socio-economic individuals with limited access to devices capable of completing the survey.

Furthermore, the results are in terms of relative importance to the most important feature. No distinction between degree of importance was made with respect to effect size for females or males. Thus, no conclusion about how each feature meaningfully contributes to clinical levels of depression, anxiety, or stress may be drawn.

With respect to the models, unweighted models (i.e., AdaBoost, Gradient Boosting, or HistGradientBoost) tended to perform between 2-to-6% better than unweighted models for the demographic predictors of depression (Table S4; Fig. S4), stress (Table S5; Fig. S5), and anxiety (Table S6; Fig. S6). Although modestly, these models artificially increased Ensemble accuracy, thus highlighting the importance of weighting in classification models. This is clear in the confusion matrices for demographic predictors, where positive cases were predicted with much higher accuracy than negative cases (Figs. S7, 8, 9).

Future research should consider both effect size and important feature splits. The impact of each feature, and how those features split for various individuals (e.g., heterosexual vs. homosexual) would yield more inclusive and granular results, thus serving a wider population. Given the age groups, future research would further parse results by age range to understand at which point emphasizing language learning about depression, anxiety, or stress independently would be more advantageous.

Conclusion

This study extracted important features from machine model predictions of depression, anxiety, and stress for females and males to better understand which features best described these mental health indicators. Gender differences suggest that the more granularly females can distinguish between anxiety and stress, the more likely they will be able to parse these from depression. For females, a granular understanding of personality, especially on agreeableness and emotional stability, might enhance their ability to describe and understand the sources of depression, anxiety, and stress.

Males, by contrast, would benefit from a breadth of descriptors. The results demonstrated that for males, DASS-21 subscales aligned with each mental health indicator. Thus, a greater emphasis on vocabulary to describe each indicator would be beneficial for males. In terms of demographic variables, males would benefit from understanding the interaction between emotional stability, family, and sexual orientation.

A clinically aligned vocabulary would provide deeper personal insight, and enhance communication with friends, family, significant others, and mental health experts. As one beloved neighbor said, anything that is human is mentionable, and anything mentionable is more manageable. At a time when Americans are reporting higher levels of depression, anxiety, and stress, there is great utility in having the language to mention it, simply that one can manage it.

Table 1

DASS-21 Scales, Subscales, Associated Questions, and Root Vocabulary

Item no.	**SCALE** *Subscale* Item	Root Vocabulary
	DEPRESSION	
	Anhedonia	Anhedonia
3	I couldn't seem to experience any positive feeling at all	Positive feeling
	Devaluation of life	Devalue
38	I felt that life was meaningless	Meaningless
	Dysphoria	Dysphoria
26	I felt down-hearted and blue	Down-hearted, blue
	Hopelessness	Hopeless
10	I felt that I had nothing to look forward to	Look forward to
	Inertia	Inertia
42	I found it difficult to work up the initiative to do things	Initiative
	Lack of interest / involvement	Interest, involvement
31	I was unable to become enthusiastic about anything	Enthusiasm
	Self-deprecation	Self-deprecation
17	I felt that I wasn't worth much as a person	Worth
	ANXIETY	Anxiety
	Autonomic arousal	
2	I was aware of dryness of my mouth	Dry mouth
4	I experienced breathing difficulty (e.g., excessively rapid breathing, breathlessness in the absence of physical exertion)	Difficulty breathing
25	I was aware of the action of my heart in the absence of physical exertion (e.g., sense of heart rate increase, heart missing a beat)	Heartbeat
	Situational anxiety	
40	I was worried about situations in which I might panic and make a fool of myself	Panic, fool
	Skeletal musculature effects	
41	I experienced trembling (e.g., in the hands)	Trembling
	Subjective experience of anxious affect	
20	I felt scared without any good reason	Scared
28	I felt I was close to panic	Panic
	STRESS	Stress
	Difficulty relaxing	Relaxing
8	I found it difficult to relax	
22	I found it hard to wind down	Wind-down
	Easily upset / agitated	Upset, agitated
11	I found myself getting agitated	
	Impatient	Impatient
35	I was intolerant of anything that kept me from getting on with what I was doing	Intolerant
	Irritable / over-reactive	Irritable, over-react
6	I tended to over-react to situations	
18	I felt I was rather touchy	Touchy
	Nervous arousal	
12	I felt that I was using a lot of nervous energy	Nervous

Note: Item numbers are based on the DASS-42 survey items included in the Answers to the Depression Anxiety Stress Scales data set. The items in this table are those related to the DASS-21 only.

Table 2

Descriptive Statistics for Age Group, Religion, Depression, Anxiety, and Stress by Gender

	Female		Male	
Age Group (years)	n	%	n	%
Adolescent (13-18)	7815	25.92	2044	23.55
Young adult (19-24)	14355	47.60	3645	42
Adult (25-44)	6992	23.19	2367	27.27
Middle aged (45-65)	943	3.13	566	6.52
Aged (66-79)	48	.16	53	.61
80 and over	3	.01	4	.05
Religion				
Religious	4243	14.07	2745	31.63
Not Religious	25913	85.93	5934	68.37
Depression				
Low to mild	9243	30.65	3035	34.97
Moderate to severe	20913	69.35	5644	65.03
Anxiety				
Low to mild	8862	29.39	3636	41.89
Moderate to severe	21294	70.61	5043	58.11
Stress				
Low to mild	11629	38.56	4567	52.62
Moderate to severe	18527	61.44	4112	47.38

Table 3

DASS-21 Clinical Cutoff Scores for Emotional States

Severity Level	Depression	Anxiety	Stress
Normal	0-9	0-7	0-14
Mild	10-13	8-9	15-18
Moderate	14-20	10-14	19-25
Severe	21-27	15-19	26-33
Extremely Severe	≥ 28	≥ 20	≥ 34

Table 4

Descriptive Statistics for Age by Gender

Age (years)	n	M	SD	Min	Max
Gender					
Female	30156	22.99	7.89	13	85
Male	8679	25.05	10.37	13	89

Note: M = mean; SD = standard deviation; Min = minimum; Max = maximum

Table 5

Ensemble Model Results for DASS-21 Subscales as Predictors of Depression, Anxiety, and Stress

	Accuracy	Precision	Recall	F1 Score
Depression				
Female	1	1	1	1
Male	.99	.99	.99	.99
Anxiety				
Female	1	1	1	1
Male	1	1	1	1
Stress				
Female	1	1	1	1
Male	1	1	1	1

Note: All reported scores are based on weighted averages

Table 6

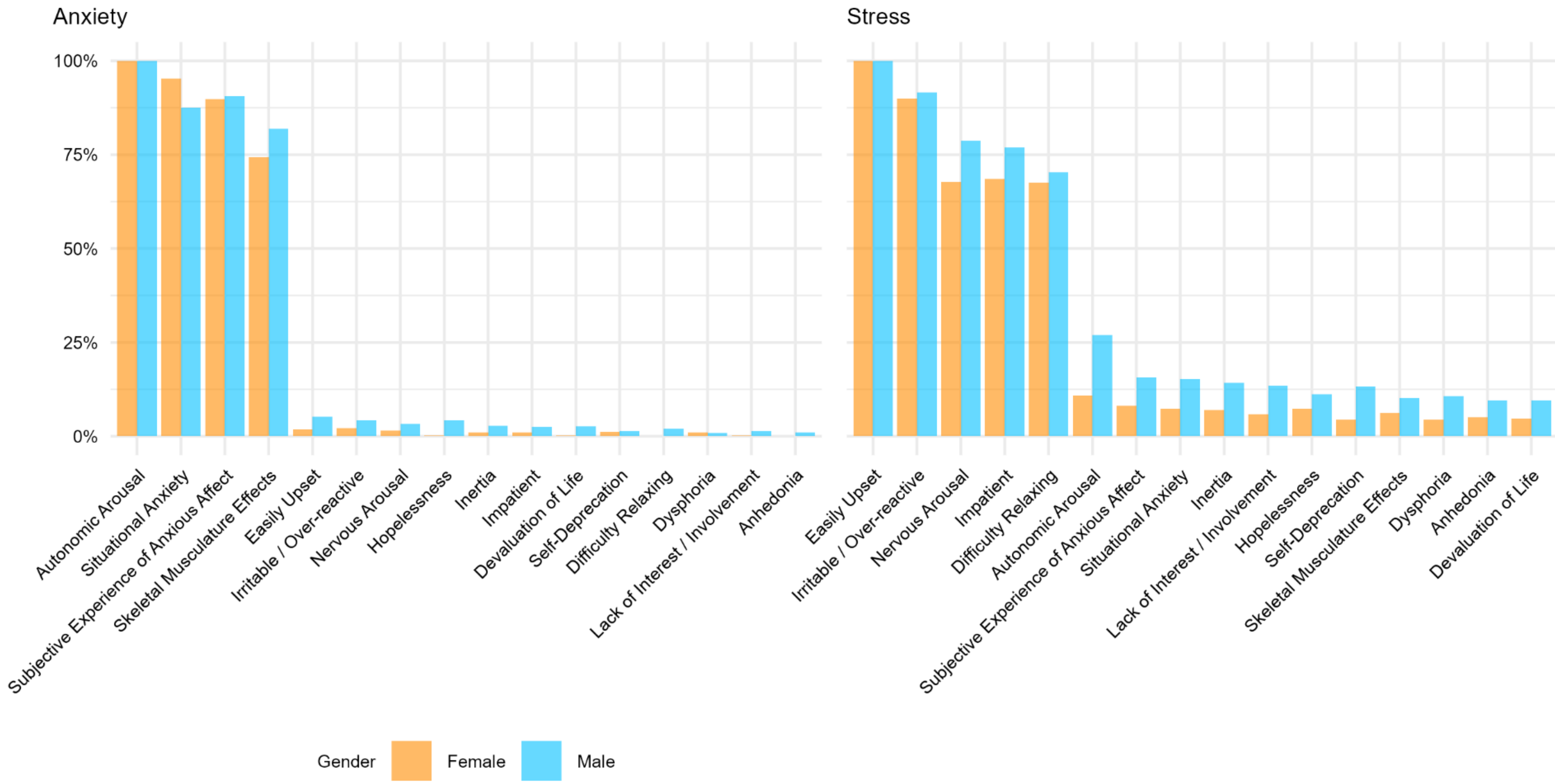
Ensemble Model Feature Important for DASS-21 Subscales

SCALE Subscale	Female Split Value	 Importance Score	 Subscale Importance	SCALE Subscale	Male Split Value	 Importance Score	 Relative Importance
DEPRESSION				**DEPRESSION**			
Easily Upset	7	476.68	1.00	Inertia	3	126.15	1.00
Impatient	1	456.00	.96	Hopelessness	1	122.21	.97
Irritable / Over-reactive	5	409.68	.86	Self-Deprecation	3	112.87	.89
Autonomic Arousal	1	401.67	.84	Lack of Interest / Involvement	1	98.29	.78
Subjective Experience of Anxious Affect	3	350.52	.74	Dysphoria	3	92.04	.73
Dysphoria	3	310.74	.65	Devaluation of Life	1	85.42	.68
Nervous Arousal	3	306.67	.64	Anhedonia	1	83.79	.66
Situational Anxiety	3	284.34	.60	Autonomic Arousal	1	80.53	.64
Inertia	1	281.19	.59	Irritable / Over-reactive	5	77.21	.61
Lack of Interest / Involvement	1	265.28	.56	Easily Upset	5	73.89	.59
Anhedonia	1	261.59	.55	Situational Anxiety	3	60.18	.48
Difficulty Relaxing	1	260.33	.55	Impatient	3	60.17	.48
Self-Deprecation	1	257.59	.54	Subjective Experience of Anxious Affect	3	54.19	.43
Devaluation of Life	1	237.61	.50	Difficulty Relaxing	3	49.19	.39
Skeletal Musculature Effects	1	235.67	.49	Nervous Arousal	1	47.67	.38
Hopelessness	1	221.77	.47	Skeletal Musculature Effects	1	43.54	.35
ANXIETY				**ANXIETY**			
Autonomic Arousal	1	139.90	1.00	Autonomic Arousal	3	132.03	1.00
Situational Anxiety	1	133.21	.95	Subjective Experience of Anxious Affect	1	119.69	.91
Subjective Experience of Anxious Affect	1	125.64	.90	Situational Anxiety	1	115.67	.88
Skeletal Musculature Effects	1	104.03	.74	Skeletal Musculature Effects	1	108.14	.82
Irritable / Over-reactive	7	3.00	.02	Easily Upset	3	6.87	.05
Easily Upset	5	2.53	.02	Hopelessness	1	5.67	.04
Nervous Arousal	1	2.01	.01	Irritable / Over-reactive	3	5.51	.04
Self-Deprecation	5	1.67	.01	Nervous Arousal	1	4.36	.03
Impatient	1	1.50	.01	Inertia	3	3.67	.03
Inertia	1	1.50	.01	Devaluation of Life	1	3.51	.03
Dysphoria	5	1.34	.01	Impatient	1	3.35	.03
Devaluation of Life	1	.33	.00	Difficulty Relaxing	3	2.68	.02
Hopelessness	1	.33	.00	Lack of Interest / Involvement	1	1.84	.01
Lack of Interest / Involvement	1	.33	.00	Self-Deprecation	1	1.84	.01
Difficulty Relaxing	5	.00	.00	Anhedonia	1	1.34	.01
Anhedonia	1	.00	.00	Dysphoria	1	1.18	.01
STRESS				**STRESS**			
Easily Upset	5	1025.23	1.00	Easily Upset	7	578.75	1.00
Irritable / Over-reactive	3	923.04	.90	Irritable / Over-reactive	5	530.19	.92
Impatient	1	701.89	.68	Nervous Arousal	3	455.32	.79
Nervous Arousal	1	694.67	.68	Impatient	3	444.86	.77
Difficulty Relaxing	1	692.96	.68	Difficulty Relaxing	3	406.43	.70
Autonomic Arousal	5	111.68	.11	Autonomic Arousal	1	155.84	.27
Subjective Experience of Anxious Affect	3	83.00	.08	Subjective Experience of Anxious Affect	1	90.69	.16
Situational Anxiety	3	75.34	.07	Situational Anxiety	3	88.17	.15
Hopelessness	1	75.17	.07	Inertia	3	82.19	.14
Inertia	1	70.84	.07	Lack of Interest / Involvement	1	77.84	.13
Skeletal Musculature Effects	1	64.17	.06	Self-Deprecation	1	77.02	.13
Lack of Interest / Involvement	1	60.33	.06	Hopelessness	1	64.67	.11
Anhedonia	1	51.67	.05	Dysphoria	1	61.52	.11
Devaluation of Life	1	48.67	.05	Skeletal Musculature Effects	5	58.84	.10
Dysphoria	5	45.17	.04	Devaluation of Life	5	55.67	.10
Self-Deprecation	1	45.00	.04	Anhedonia	3	55.17	.10

Note: Relative importance was calculated by dividing each importance score by the highest score

Figure 1

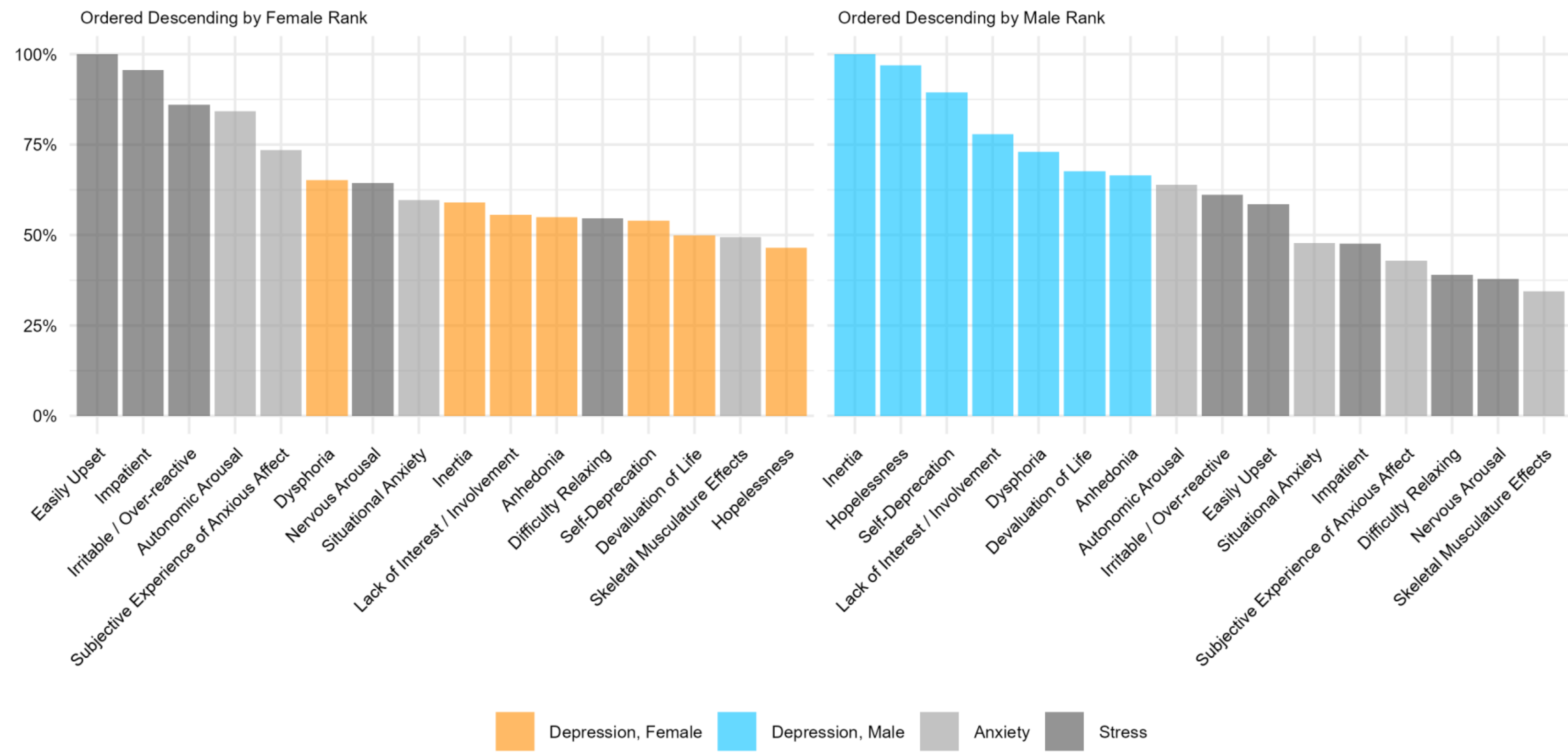
Top DASS-21 Subscale Predictors of Anxiety and Stress



This graphic shows relative feature importance for females and males with respect to anxiety (left) and stress (right). Subscales for anxiety and stress predictably align with their respective scales, suggesting the measures have strong discriminant validity. Feature importance for anxiety and stress are similar for both males and females.

Figure 2

Top DASS-21 Subscale Predictors of Depression



The chart on the left shows relative feature importance scores in descending order for females, whereas the chart on the right shows scores in descending order for males. This highlights how important features for females and males differ. For example, features more predictive of anxiety and stress drive depression scores for females, while the associated subscales for depression best predict depression for males.

Table 7

Ensemble Model Results for Demographic Variables as Predictors of Depression, Anxiety, and Stress

	Accuracy	Precision	Recall	F1 Score
Depression				
Female	.76	.76	.76	.76
Male	.76	.76	.76	.76
Anxiety				
Female	.74	.73	.74	.74
Male	.72	.72	.72	.72
Stress				
Female	.75	.75	.75	.75
Male	.74	.74	.74	.74

Note: All reported scores are based on weighted averages

Table 8

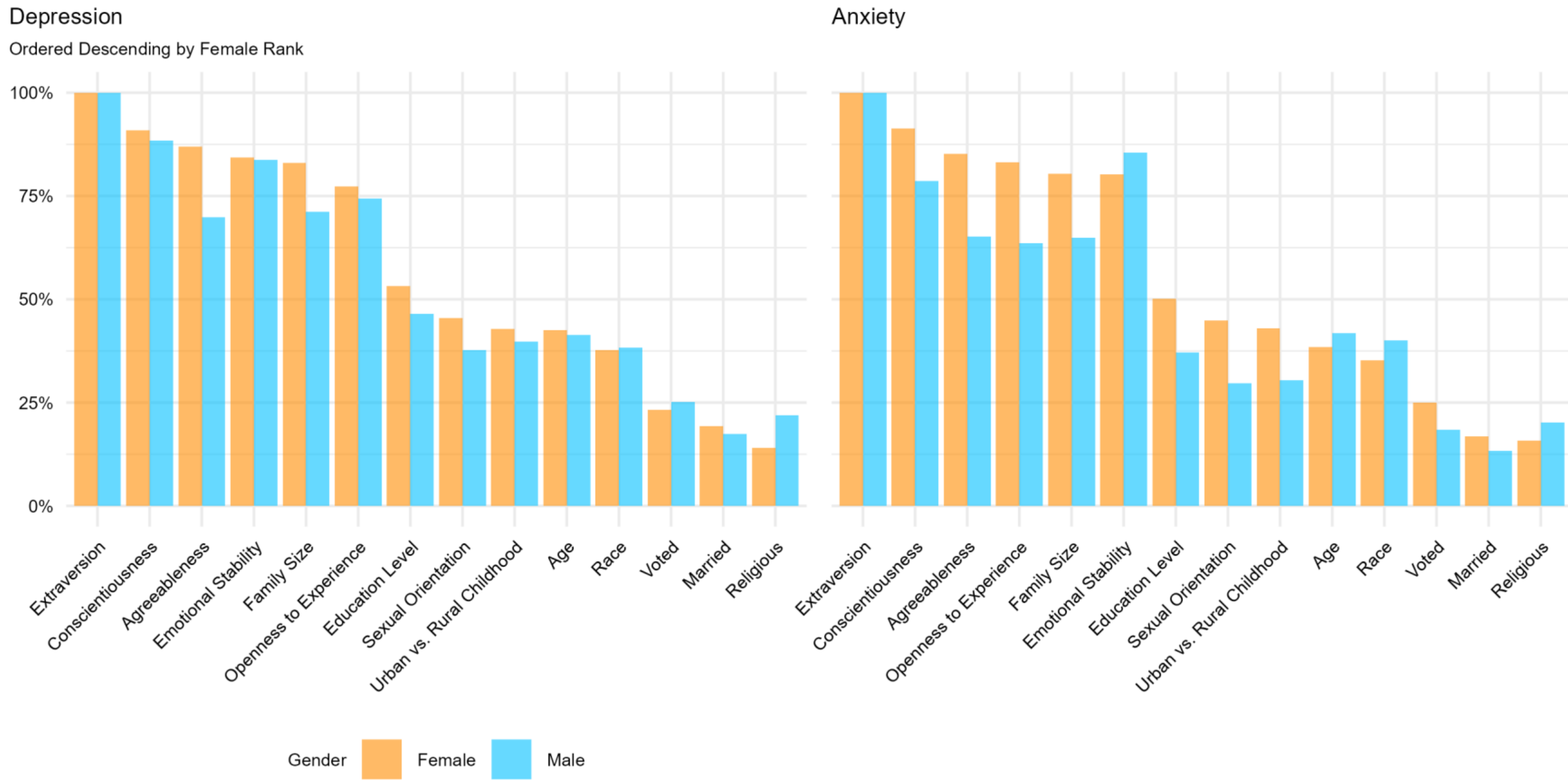
Ensemble Model Feature Important for Demographic Variables

SCALE Demographic Variable	Female Split Value	 Importance Score	 Relative Importance	SCALE Demographic Variable	Male Split Value	 Importance Score	 Relative Importance
DEPRESSION				**DEPRESSION**			
Extraversion	3.83	314.29	1.00	Extraversion	3.17	167.70	1.00
Conscientiousness	4.75	285.75	.91	Conscientiousness	4.25	148.34	.88
Agreeableness	4.25	273.47	.87	Emotional Stability	3.25	14.42	.84
Emotional Stability	3.75	264.92	.84	Openness to Experience	4.25	124.87	.74
Family Size	3.50	261.03	.83	Family Size	1.50	119.41	.71
Openness to Experience	4.75	243.14	.77	Agreeableness	4.25	117.24	.70
Education Level	2.50	167.32	.53	Education Level	2.50	77.85	.46
Sexual Orientation	1.50	142.99	.45	Age Group	1.50	69.30	.41
Urban vs. Rural Childhood	2.50	134.41	.43	Urban vs. Rural Childhood	1.50	66.76	.40
Age Group	1.50	133.51	.42	Race	5.50	64.19	.38
Race	5.50	118.56	.38	Sexual Orientation	.50	63.31	.38
Voted	1.50	72.87	.23	Voted	1.50	42.12	.25
Married	1.50	60.77	.19	Religious	.50	36.72	.22
Religious	.50	44.32	.14	Married	1.50	29.09	.17
ANXIETY				**ANXIETY**			
Extraversion	3.50	635.39	1.00	Extraversion	2.83	75.10	1.00
Conscientiousness	4.25	580.54	.91	Emotional Stability	2.75	64.18	.85
Agreeableness	4.25	541.78	.85	Conscientiousness	4.25	59.10	.79
Openness to Experience	5.25	528.29	.83	Agreeableness	4.75	48.92	.65
Family Size	2.50	51.73	.80	Family Size	2.50	48.76	.65
Emotional Stability	2.25	509.79	.80	Openness to Experience	4.25	47.72	.64
Education Level	2.50	318.72	.50	Age Group	.50	31.42	.42
Sexual Orientation	.50	285.20	.45	Race	5.50	30.12	.40
Urban vs. Rural Childhood	2.50	273.23	.43	Education Level	2.50	27.89	.37
Age Group	.50	244.44	.38	Urban vs. Rural Childhood	2.50	22.85	.30
Race	5.50	223.45	.35	Sexual Orientation	1.50	22.30	.30
Voted	1.50	158.85	.25	Religious	.50	15.15	.20
Married	1.50	106.95	.17	Voted	1.50	13.86	.18
Religious	.50	99.98	.16	Married	1.50	9.97	.13
STRESS				**STRESS**			
Extraversion	3.50	635.39	1.00	Emotional Stability	2.75	97.89	1.00
Conscientiousness	4.25	580.54	.91	Family Size	2.50	79.91	.82
Agreeableness	4.25	541.78	.85	Sexual Orientation	.50	56.32	.58
Openness to Experience	5.25	528.29	.83	Agreeableness	4.25	53.83	.55
Family Size	2.50	510.73	.80	Education Level	2.50	48.02	.49
Emotional Stability	2.25	509.79	.80	Age Group	.50	38.69	.40
Education Level	2.50	318.72	.50	Openness to Experience	5.75	32.02	.33
Sexual Orientation	.50	285.20	.45	Conscientiousness	4.25	30.17	.31
Urban vs. Rural Childhood	2.50	273.23	.43	Extraversion	3.50	19.10	.20
Age Group	.50	244.44	.38	Voted	1.50	15.02	.15
Race	5.50	223.45	.35	Religious	.50	14.25	.15
Voted	1.50	158.85	.25	Urban vs. Rural Childhood	2.50	14.04	.14
Married	1.50	106.95	.17	Race	5.50	11.37	.12
Religious	.50	99.98	.16	Married	1.50	6.70	.07

Note: Relative importance was calculated by dividing each importance score by the highest score

Figure 3

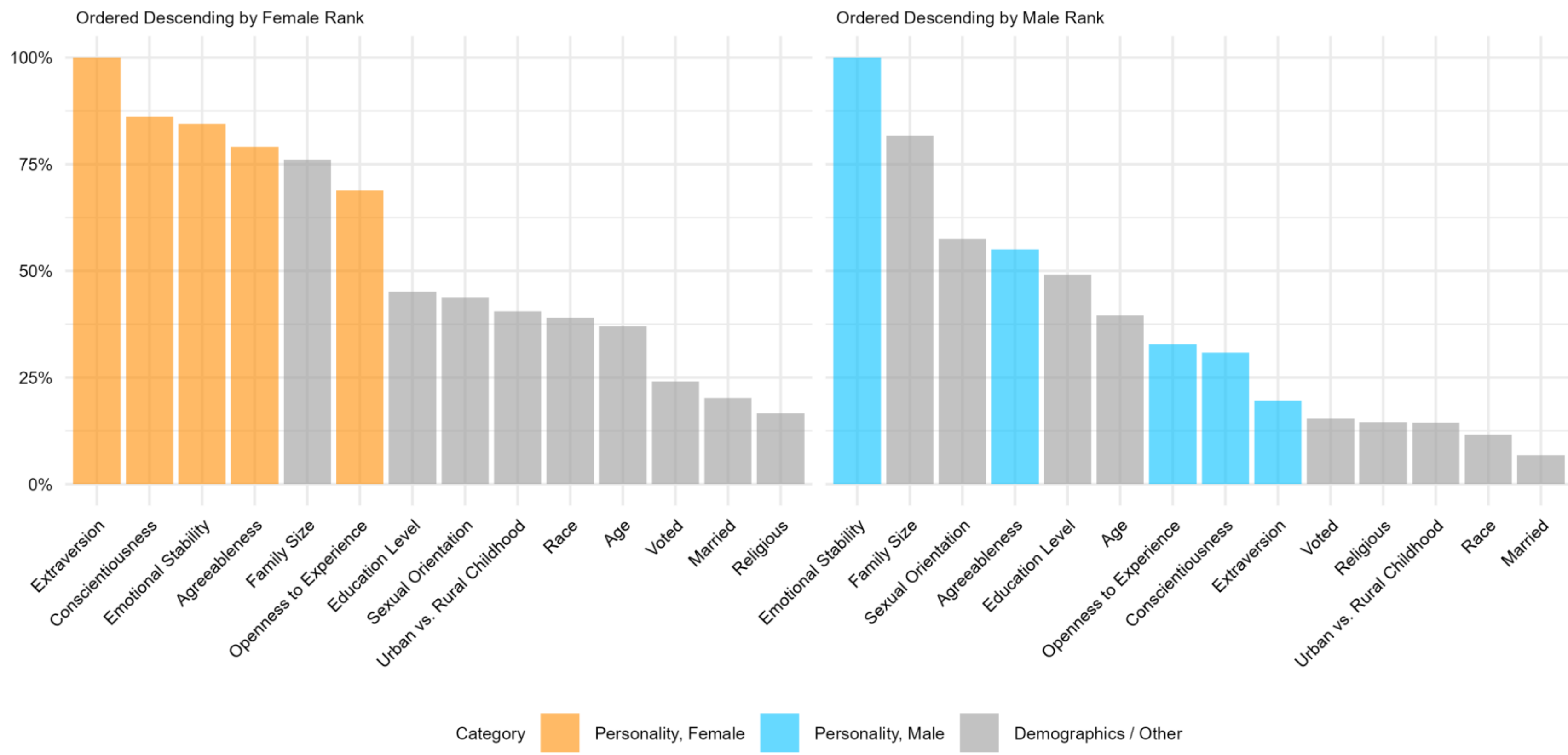
Top Demographic Predictors of Depression and Anxiety



This graphic shows relative feature importance for females and males with respect to depression (left) and anxiety (right). Features are arranged in descending order by importance for females. While demographic feature importance is relatively similar for females and males, emotional stability is more important for males than females for predicting anxiety.

Figure 4

Top Demographic Predictors of Stress



The chart on the left shows relative feature importance scores in descending order for females, whereas the chart on the right shows importance scores in descending order for males. This highlights how important features for females and males differ. For example, personality features drive stress predictors for females, while non-personality features like family size, sexual orientation, and education level are of greater importance for males.

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